

Characterization of the Thermal Properties of Free Flaps

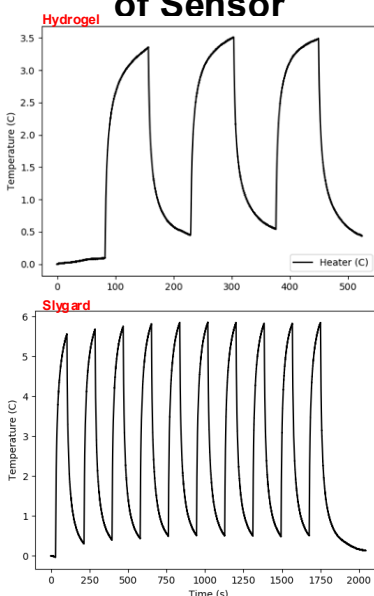
Objectives:

- To validate pig trial flap data with a benchtop experiment
- To find a correlation between thermal isotropy and conductivity
- To optimize device sensitivity to assess flap perfusion

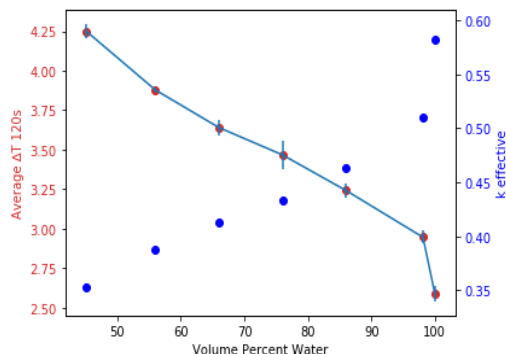
Results:

- The overall sensitivity of the device by modulating the water content
- Thermal conductivity and ΔT are inversely correlated
- The drift in the hydrogel experiments is a result of bad contact. (the device is peeling off)
- The drift experiment with the skin phantom (sylgard), we saw very little drift in first 3 cycles but in all cycles ΔT hasn't been affected.

Drift Characterization of Sensor

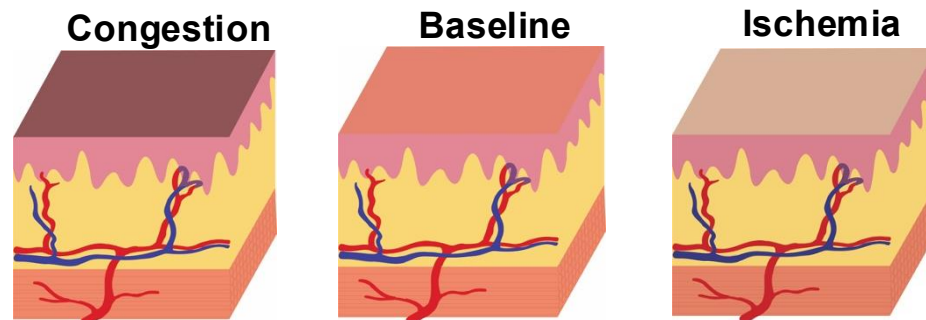


Hydrogel – Volume Percent Water Loss



Background & Technical Approach:

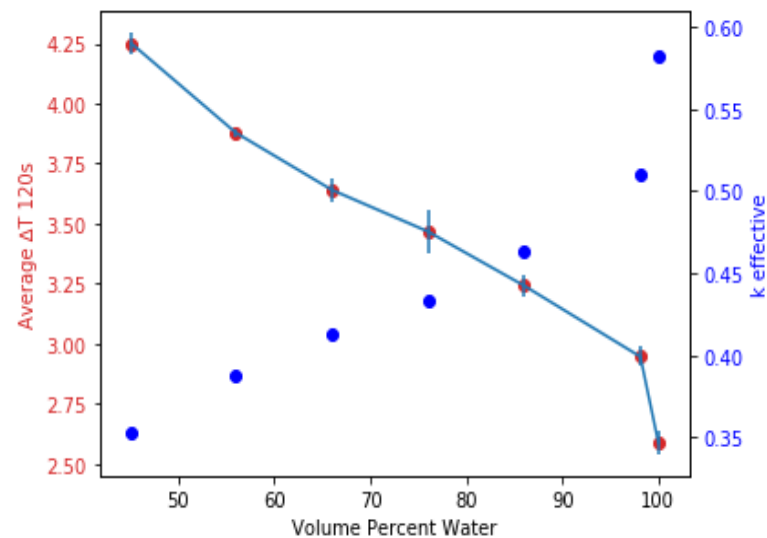
- Skin flap: skin and subcutaneous tissue that survives based on its own blood supply.
- Types of skin flaps and surgical techniques may vary depending on the severity of the wound.
- Skin flap viability is commonly monitored either surgically, which may cause infections, or StO_2 oximetry, which may yield inaccurate results



Working Principle of Device & Materials:

- In comparison to previous flow device, this one has an extra sensor located in the heater; in this case we're using the central sensor for trials.
- To validate pig data via a benchtop experiment the material 'Hydrogel' was used.
- Hydrogel is a 'sponge-like' material that, in this case, is used to measure water loss with the device. The experiment imitates the water content in a skin model, correlated to ΔT .

Hydrogel – Volume Percent Water Loss



Drift Characterization of Sensor

